SFU SCHOOL OF COMPUTING SCIENCE

RL Agent: Maze Solver

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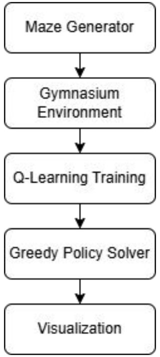
Purpose

This project explores how a reinforcement learning agent can learn to navigate and solve randomly generated mazes. We train a Q-learning agent inside a custom maze environment so it can discover an efficient path from the start to the goal through repeated interactions.

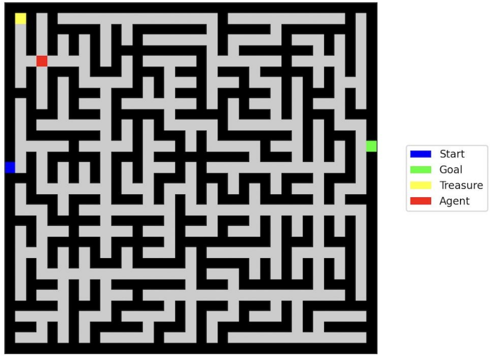
Project Overview

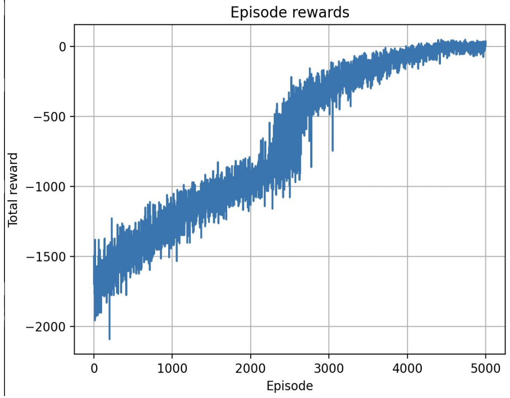
We built a complete RL system that learns to solve procedural mazes. Each run generates a new maze using Prim's algorithm. The agent interacts with a Gymnasium-style environment and learns an optimal navigation strategy using Q-learning.
After training, the agent solves the maze using a greedy policy, and its path is visualized with start, goal, and agent markers.

Pipeline Overview



Visualization





Methods

We generate perfect mazes with Prim's algorithm and place a treasure the agent must collect before exiting. The environment gives simple rewards for movement, walls, treasure pickup, and reaching the exit. Our agent uses tabular Q-learning with $\alpha = 0.1$, $\gamma = 0.95$, and an ε-greedy policy that decays from 1.0 to 0.05. After training, we evaluate the agent using a greedy policy. For comparison, we use Breadth-First Search, which always returns the shortest path and serves as our optimal baseline.

Results

After training, the Q-learning agent finds routes that match the BFS path length, despite not having access to the maze layout. Early episodes are inefficient, but the agent quickly learns to grab the treasure and reach the exit cleanly. Matching the BFS baseline shows that the learned policy is near-optimal and that tabular Q-learning can solve these mazes from experience alone.

Github:
https://github.com/CMPT310-final-proj/rl-maze-solver.git